

Introduction to Data Analytics: SGA-1

Assignment Type: Individual

Submission format: Single PDF file

Learning Objectives

By completing this assignment, students will be able to:

1. Differentiate between Descriptive, Diagnostic, Predictive, and Prescriptive analytics.
 2. Classify data based on structure, source, and nature (qualitative vs quantitative).
 3. Identify appropriate data sources for real-world business problems.
 4. Analyze challenges arising from unstructured, proxy, and secondary data.
 5. Apply analytical thinking to volatile and uncertain environments.
 6. Understand ethical, technical, and diagnostic limitations of data-driven decision-making.
 7. Design end-to-end analytics workflows across industries.
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Instructions to Students

- Answer **all questions** clearly and concisely.
 - Support your answers with relevant examples, concepts, and terminology from the course.
 - Where required, justify your choices logically.
 - Your submission must be a **single PDF file** containing:
 - Student Name
 - Email Address
 - Roll Number
 - **Clearly numbered solutions** corresponding to each question
 - Ensure clarity, structure, and readability.
 - Plagiarism or use of AI-generated content without understanding may lead to penalties.
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Questions

Question 1: Analytics Types Across Business Investigation

A multinational retail company observes a sudden decline in quarterly profits.

- (a) Identify which type of analytics (Descriptive, Diagnostic, Predictive, Prescriptive) should be applied at each stage of investigation.
- (b) Justify your choices with examples of techniques used at each stage.

Question 2: Social Media Sentiment Analysis

A company wants to analyze customer sentiment using social media data.

- (a) Identify the data type and data source
- (b) Discuss the major challenges in analyzing this data
- (c) Suggest suitable analytics technique.

Question 3: The Maturity Paradox

An organization has mastered Descriptive and Diagnostic analytics with very high accuracy. However, its Predictive models fail due to a sudden **Black Swan structural shift** in the industry.

Should the organization:

- Invest more in Prescriptive Analytics, or
- Revert to Qualitative data gathering?

Justify your answer using the relationship between **data structure and environmental volatility**.

Question 4: Multi-Source Integration Challenge

A global retail chain wants to predict localized supply chain disruptions using:

- Internal / Structured: POS records
- Internal / Semi-Structured: XML logistics logs
- External / Unstructured: Satellite imagery and social media sentiment

(a) Design a data ingestion strategy that prioritizes **Primary data** over Secondary data to reduce bias.

(b) Identify which data source has the highest **noise-to-signal ratio** and explain why.

Question 5: Diagnostic Analytics in Manufacturing

An aerospace manufacturer faces a high defect rate and has access to:

- Structured sensor logs
- Unstructured handwritten engineer notes
- Diagnostic X-ray scans

Design a **Diagnostic Analytics workflow** integrating all three data sources.

Identify the **dependent variable** and **independent variables** in this analysis.

Expectations from Students

- Demonstrate **conceptual clarity** rather than memorization.
- Use correct analytics terminology consistently.
- Show ability to **reason under uncertainty** and explain trade-offs.
- Present structured, well-organized answers.
- Apply concepts to **real-world scenarios**, not just textbook definitions.
- Ethical awareness and critical thinking will be positively evaluated.